



UNIVERSITE DE FIANARANTSOA

ECOLE NATIONALE D'INFORMATIQUE

Rapport de Projet Routage IP

Mention : Informatique

Parcours : Informatique Générale

INTITULE :

Etude et Redistribution de Protocole de Routage RIP version 2 et OSPF

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Matricule : 2205 H-F

Matricule : 2267 H-F

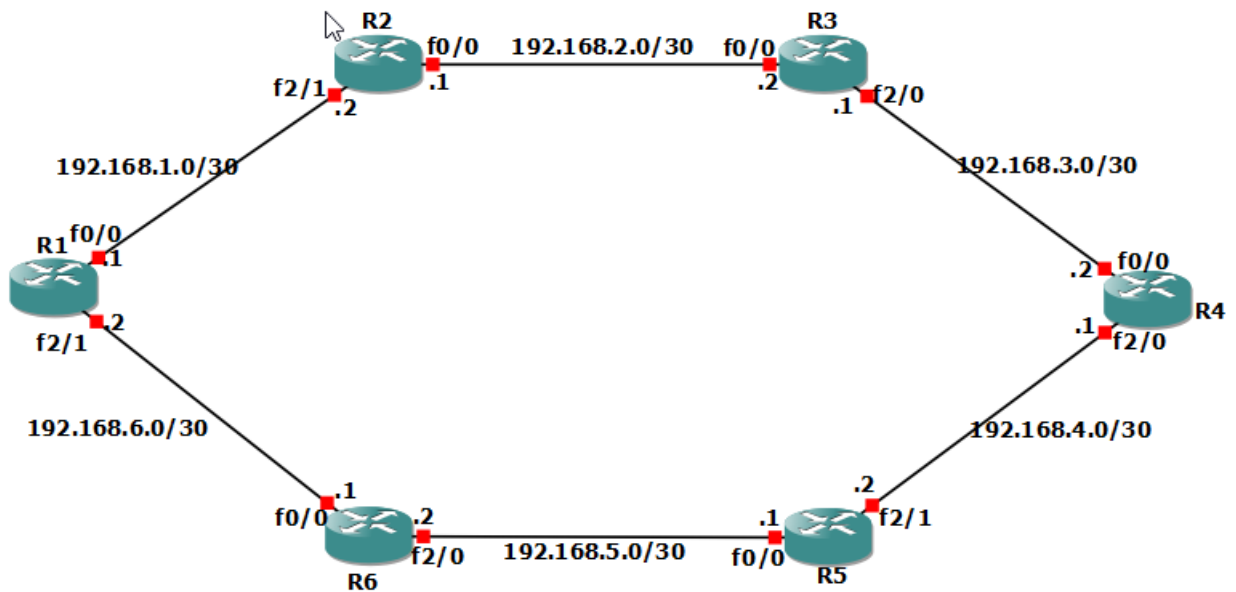
Matricule : 2272 H-F

Matricule : 2303 H-F

Enseignant :

- Mr RAZAFINDRAMONJA Clément Aubert

TOPOLOGIE DE LA PARTIE RIP :



❖ CONFIGURATION DES ROUTEURS

Configuration de chaque routeur et activation de RIPv2 , et on va le commencer par le routeur R1 .
Tous les routeurs sont tous configurés selon leurs cas, ici on parle d'abord du **RIP**.

CONFIGUTRATION DU ROUTEUR R1 :

La configuration du routeur est présentées ci-dessous :

```

R1 R2 R3 R4 R5 R6 R7 R8 R9
*Apr 30 09:00:32.427: %LINK-5-CHANGED: Interface Ethernet1/4, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface Ethernet1/5, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface Ethernet1/6, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface Ethernet1/7, changed state to administratively down
*Apr 30 09:00:32.431: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state to administratively down
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#int f0/0
R1(config-if)#ip add 192.168.1.1 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#
*Apr 30 09:01:41.663: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:01:42.663: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit
R1(config)#int f2/1
R1(config-if)#ip add 192.168.6.2 255.255.255.252
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#
*Apr 30 09:02:21.175: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:02:22.175: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R1(config)#router rip
R1(config-router)#version 2
R1(config-router)#network 192.168.1.0
R1(config-router)#network 192.168.6.0
R1(config-router)#no auto-summary
R1(config-router)#end
R1#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
*Apr 30 09:57:58.827: %SYS-5-CONFIG_I: Configured from console by console
[confirm]
Building configuration...
[OK]
R1#
```

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CONFIGURATION DU ROUTEUR R2 :

```

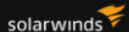
R1 R2 R3 R4 R5 R6 R7 R8 R9
*Apr 30 09:04:18.127: %LINK-5-CHANGED: Interface Ethernet1/5, changed state to administratively down
*Apr 30 09:04:18.139: %LINK-5-CHANGED: Interface Ethernet1/6, changed state to administratively down
*Apr 30 09:04:18.143: %LINK-5-CHANGED: Interface Ethernet1/7, changed state to administratively down
*Apr 30 09:04:18.167: %LINK-5-CHANGED: Interface FastEthernet2/0, changed state to administratively down
*Apr 30 09:04:18.171: %LINK-5-CHANGED: Interface FastEthernet2/1, changed state to administratively down
R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#int f0/0
R2(config-if)#ip add 192.168.2.1 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#int
*Apr 30 09:06:38.415: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:06:39.415: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R2(config)#int f2/1
R2(config-if)#ip add 192.168.1.2 255.255.255.252
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#
*Apr 30 09:07:16.795: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:07:17.795: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 192.168.1.0
R2(config-router)#network 192.168.2.0
R2(config-router)#end
R2#
*Apr 30 09:57:46.887: %SYS-5-CONFIG_I: Configured from console by console
R2#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R2#
```

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CONFIGURATION DU ROUTEUR R3 :

```

R1 R2 R3 x R4 R5 R6 R7 R8 R9
*Apr 30 09:09:20.327: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/4, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/5, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/6, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/7, changed state to down
*Apr 30 09:09:20.331: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#int f0/0
R3(config-if)#ip add 192.168.2.2 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
R3(config)#
*Apr 30 09:10:11.075: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:10:12.075: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R3(config)#int f2/0
R3(config-if)#ip add 192.168.3.1 255.255.255.252
R3(config-if)#no shutdown
R3(config-if)#exit
*Apr 30 09:10:53.275: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:10:54.275: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R3(config-if)#exit
R3(config)#router rip
R3(config-router)#version 2
R3(config-router)#network 192.168.2.0
R3(config-router)#network 192.168.3.0
R3(config-router)#no auto-summary
R3(config-router)#end
R3#
*Apr 30 09:57:30.567: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R3#
```



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CONFIGURATION DU ROUTEUR R4 :

```
R1 R2 R3 R4 x R5 R6 R7 R8 R9 + - □ ×

*Apr 30 09:12:20.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int f0/0
R4(config-if)#ip add 192.168.3.2 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:08.539: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:16:09.539: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config)#int f2/0
R4(config-if)#ip add 192.168.4.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:58.587: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:16:59.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R4(config)#int f2/1
R4(config-if)#ip add 172.16.10.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:17:49.963: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:17:50.963: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
R4(config-router)#

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```

CONFIGURATION DU ROUTEUR R5 :

```

R1 R2 R3 R4 R5 x R6 R7 R8 R9
*Apr 30 09:22:06.755: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/6, changed state to down
*Apr 30 09:22:06.763: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/7, changed state to down
*Apr 30 09:22:06.771: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
*Apr 30 09:22:06.783: %LINEPROTO-5-UPDOWN: Line pro
R5#cotocol on Interface FastEthernet2/1, changed state to down
R5#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R5(config)#int f2/1
R5(config-if)#ip add 192.168.4.2 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#exit
R5(config)#
*Apr 30 09:22:54.487: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:22:55.487: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R5(config)#int f0/0
R5(config-if)#ip add 192.168.5.1 255.255.255.252
R5(config-if)#no shutdown
R5(config-if)#
*Apr 30 09:23:38.943: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:23:39.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R5(config-if)#exit
R5(config)#router rip
R5(config-router)#version 2
R5(config-router)#no auto-summary
R5(config-router)#network 192.168.4.0
R5(config-router)#network 192.168.5.0
R5(config-router)#end
R5#wr
*Apr 30 09:58:08.607: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R5#
```

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CONFIGURATION DU ROUTEUR R6 :

```

R1 R2 R3 R4 R5 R6 x R7 R8 R9 | + - □ ×
*Apr 30 09:24:49.939: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/4, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/5, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/6, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface Ethernet1/7, changed state to down
*Apr 30 09:24:49.943: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R6#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R6(config)#int f2/0
R6(config-if)#ip add 192.168.5.2 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
R6(config)#
*Apr 30 09:25:40.283: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:25:41.283: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R6(config)#int f0/0
R6(config-if)#ip add 192.168.6.1 255.255.255.252
R6(config-if)#no shutdown
R6(config-if)#exit
*Apr 30 09:26:35.511: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:26:36.511: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R6(config-if)#exit
R6(config)#router rip
R6(config-router)#version 2
R6(config-router)#no auto-summary
R6(config-router)#network 192.168.5.0
R6(config-router)#network 192.168.6.0
R6(config-router)#end
R6#w
*Apr 30 09:58:19.567: %SYS-5-CONFIG_I: Configured from console by console
R6#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R6#
```

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❖ OBSERVATION DES TABLES DE ROUTAGE :

Dans cette section, nous allons examiner les tables de routage afin de mieux comprendre le fonctionnement des réseaux et la circulation des données, permettant d'identifier les chemins empruntés par les paquets dans un réseau.

Table de routage de routeur R1 :

Voici la présentation de la table de routage du routeur R1 pour plus d'explication sur ce routeur. Elle est montrées ci-dessous :

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
R    10.0.0.7 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
      [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    10.0.0.8 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
      [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    10.0.0.9 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
      [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
172.16.0.0/30 is subnetted, 4 subnets
R    172.16.7.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
      [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    172.16.8.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
      [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    172.16.9.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
      [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    172.16.10.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
      [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, FastEthernet0/0
L    192.168.1.1/32 is directly connected, FastEthernet0/0
192.168.2.0/30 is subnetted, 1 subnets
R    192.168.2.0 [120/1] via 192.168.1.2, 00:00:24, FastEthernet0/0
192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/2] via 192.168.1.2, 00:00:24, FastEthernet0/0
192.168.4.0/30 is subnetted, 1 subnets
R    192.168.4.0 [120/2] via 192.168.6.1, 00:00:04, FastEthernet2/1
192.168.5.0/30 is subnetted, 1 subnets
R    192.168.5.0 [120/1] via 192.168.6.1, 00:00:04, FastEthernet2/1
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/30 is directly connected, FastEthernet2/1
L    192.168.6.2/32 is directly connected, FastEthernet2/1
R1#
```

- Passons maintenant au routeur R2.

Table de routage du routeur R2 :

Présentation de la table de routage R2, elle montre les chemins utilisés pour atteindre les différents réseaux.


```
[OK]
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
R    10.0.0.7 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    10.0.0.8 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    10.0.0.9 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
172.16.0.0/30 is subnetted, 4 subnets
R    172.16.7.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    172.16.8.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    172.16.9.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    172.16.10.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, FastEthernet2/1
L    192.168.1.2/32 is directly connected, FastEthernet2/1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/30 is directly connected, FastEthernet0/0
L    192.168.2.1/32 is directly connected, FastEthernet0/0
192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/1] via 192.168.2.2, 00:00:05, FastEthernet0/0
192.168.4.0/30 is subnetted, 1 subnets
R    192.168.4.0 [120/2] via 192.168.2.2, 00:00:05, FastEthernet0/0
192.168.5.0/30 is subnetted, 1 subnets
R    192.168.5.0 [120/2] via 192.168.1.1, 00:00:09, FastEthernet2/1
192.168.6.0/30 is subnetted, 1 subnets
R    192.168.6.0 [120/1] via 192.168.1.1, 00:00:09, FastEthernet2/1
R2#
```

La table de routage de R2 est bien en place pour voir les adressages vers les autres routeurs.

Tables de routage du routeur R3 :

Dans cette table de routage, elle met en évidence les réseaux directement connectés ainsi que les routes apprises via le protocole RIP, notamment à travers les routeurs R2 et R4.

```

R1 R2 R3 x R4 R5 R6 R7 R8 R9
*Apr 30 09:57:30.567: %SYS-5-CONFIG_I: Configured from console by console
R3#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
R    10.0.0.7 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R    10.0.0.8 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R    10.0.0.9 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
172.16.0.0/30 is subnetted, 4 subnets
R    172.16.7.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R    172.16.8.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R    172.16.9.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
R    172.16.10.0 [120/5] via 192.168.3.2, 00:00:20, FastEthernet2/0
192.168.1.0/30 is subnetted, 1 subnets
R    192.168.1.0 [120/1] via 192.168.2.1, 00:00:12, FastEthernet0/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/30 is directly connected, FastEthernet0/0
L    192.168.2.2/32 is directly connected, FastEthernet0/0
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/30 is directly connected, FastEthernet2/0
L    192.168.3.1/32 is directly connected, FastEthernet2/0
192.168.4.0/30 is subnetted, 1 subnets
R    192.168.4.0 [120/1] via 192.168.3.2, 00:00:20, FastEthernet2/0
192.168.5.0/30 is subnetted, 1 subnets
R    192.168.5.0 [120/2] via 192.168.3.2, 00:00:20, FastEthernet2/0
192.168.6.0/30 is subnetted, 1 subnets
R    192.168.6.0 [120/2] via 192.168.2.1, 00:00:12, FastEthernet0/0
R3#
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```

Tables de routages du routeur R4 :

Le routeur R4 regroupe les informations d'acheminement issues de deux protocoles de routage différents (RIP et OSPF). Il est le point d'interconnexion entre ces deux domaines, notamment par le routeur R7.

```
R4#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
O    10.0.0.7 [110/2] via 172.16.10.2, 00:54:10, FastEthernet2/1
O    10.0.0.8 [110/3] via 172.16.10.2, 00:40:46, FastEthernet2/1
O    10.0.0.9 [110/3] via 172.16.10.2, 00:35:26, FastEthernet2/1
172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O    172.16.7.0/30 [110/2] via 172.16.10.2, 00:54:10, FastEthernet2/1
O    172.16.8.0/30 [110/3] via 172.16.10.2, 00:40:36, FastEthernet2/1
O    172.16.9.0/30 [110/2] via 172.16.10.2, 00:54:10, FastEthernet2/1
C    172.16.10.0/30 is directly connected, FastEthernet2/1
L    172.16.10.1/32 is directly connected, FastEthernet2/1
192.168.1.0/30 is subnetted, 1 subnets
R    192.168.1.0 [120/2] via 192.168.3.1, 00:00:26, FastEthernet0/0
192.168.2.0/30 is subnetted, 1 subnets
R    192.168.2.0 [120/1] via 192.168.3.1, 00:00:26, FastEthernet0/0
192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.3.0/30 is directly connected, FastEthernet0/0
L    192.168.3.2/32 is directly connected, FastEthernet0/0
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/30 is directly connected, FastEthernet2/0
L    192.168.4.1/32 is directly connected, FastEthernet2/0
192.168.5.0/30 is subnetted, 1 subnets
R    192.168.5.0 [120/1] via 192.168.4.2, 00:00:14, FastEthernet2/0
192.168.6.0/30 is subnetted, 1 subnets
R    192.168.6.0 [120/2] via 192.168.4.2, 00:00:14, FastEthernet2/0
R4#
```

Table de routage du routeur R5 :

Il met en évidence les chemins de transmission passant par les routeurs R4 et R6 pour atteindre les autres segments du réseaux.

```

R1 R2 R3 R4 R5 x R6 R7 R8 R9 | + - □ ×
*Apr 30 09:58:08.607: %SYS-5-CONFIG_I: Configured from console by console
R5#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R5#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
R    10.0.0.7 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R    10.0.0.8 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R    10.0.0.9 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
172.16.0.0/30 is subnetted, 4 subnets
R    172.16.7.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R    172.16.8.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R    172.16.9.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
R    172.16.10.0 [120/5] via 192.168.4.1, 00:00:20, FastEthernet2/1
192.168.1.0/30 is subnetted, 1 subnets
R    192.168.1.0 [120/2] via 192.168.5.2, 00:00:14, FastEthernet0/0
192.168.2.0/30 is subnetted, 1 subnets
R    192.168.2.0 [120/2] via 192.168.4.1, 00:00:20, FastEthernet2/1
192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/1] via 192.168.4.1, 00:00:20, FastEthernet2/1
192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.4.0/30 is directly connected, FastEthernet2/1
L    192.168.4.2/32 is directly connected, FastEthernet2/1
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/30 is directly connected, FastEthernet0/0
L    192.168.5.1/32 is directly connected, FastEthernet0/0
192.168.6.0/30 is subnetted, 1 subnets
R    192.168.6.0 [120/1] via 192.168.5.2, 00:00:14, FastEthernet0/0
R5#
```

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Table de routage du routeur R6 :

Il a une fonction dans la continuité du routage au sein du domaine RIP et permet d'identifier les réseaux accessibles ainsi que les routes échangées avec les routeurs voisins (R5 et R1) .

```
R1 R2 R3 R4 R5 R6 x R7 R8 R9 | + - □ ×
R6#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
R    10.0.0.7 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R    10.0.0.8 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R    10.0.0.9 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
172.16.0.0/30 is subnetted, 4 subnets
R    172.16.7.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R    172.16.8.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R    172.16.9.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
R    172.16.10.0 [120/6] via 192.168.5.1, 00:00:28, FastEthernet2/0
192.168.1.0/30 is subnetted, 1 subnets
R    192.168.1.0 [120/1] via 192.168.6.2, 00:00:23, FastEthernet0/0
192.168.2.0/30 is subnetted, 1 subnets
R    192.168.2.0 [120/2] via 192.168.6.2, 00:00:23, FastEthernet0/0
192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/2] via 192.168.5.1, 00:00:28, FastEthernet2/0
192.168.4.0/30 is subnetted, 1 subnets
R    192.168.4.0 [120/1] via 192.168.5.1, 00:00:28, FastEthernet2/0
192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.5.0/30 is directly connected, FastEthernet2/0
L    192.168.5.2/32 is directly connected, FastEthernet2/0
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/30 is directly connected, FastEthernet0/0
L    192.168.6.1/32 is directly connected, FastEthernet0/0
R6#
R6#
```

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Dans la prochaine page, on va faire une analyse sur un routeur pour voir plus de details au routage.

❖ CONTENU D'UNE TABLE DE ROUTAGE :

• Routeur R1 :



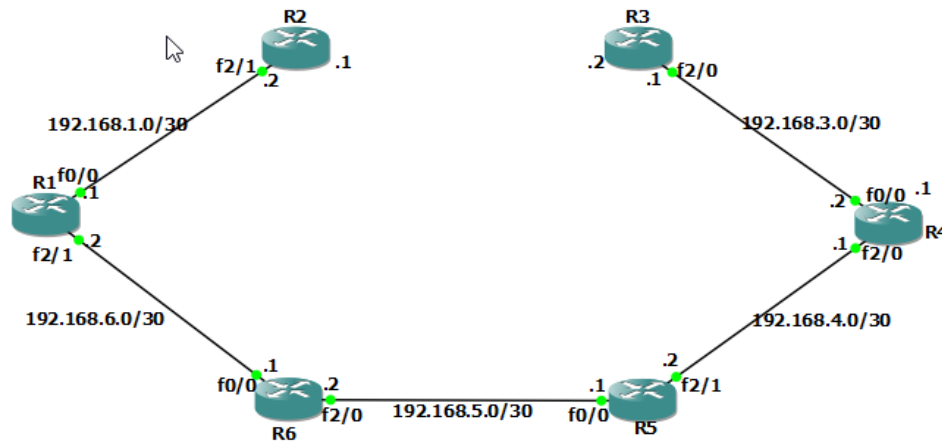
```
R1#
R1>ping 192.168.1.5
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.5, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 140/282/560 ms
R1>ping 192.168.1.6
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 360/432/504 ms
R1>ping 192.168.1.9
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.9, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 320/447/536 ms
R1>ping 192.168.1.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 556/712/864 ms
R1>ping 192.168.1.13
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.13, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 344/486/548 ms
R1>ping 192.168.1.14
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.14, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 536/723/936 ms
R1>ping 192.168.1.17
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.17, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 348/501/612 ms
R1>ping 192.168.1.18
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.18, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 196/264/344 ms
R1>ping 192.168.1.22
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.22, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 104/213/396 ms
```

La table de routage de R1 contient les informations suivantes :

- Le réseau 192.168.1.0/30 est directement connecté via l'interface FastEthernet0/0, avec une adresse de passerelle (gateway) de 192.168.1.1
- Le réseau 192.168.1.4/30 est atteint via une route apprise par le protocole de routage avec une métrique de 120 et la passerelle est 192.168.1.2 via l'interface FastEthernet0/0.
- Le réseau 192.168.1.8/30 est atteint via la même passerelle 192.168.1.2 avec une métrique de 120.
- Le réseau 192.168.1.12/30 est atteint via une autre passerelle 192.168.1.22 via l'interface Ethernet1/0 avec une métrique de 120.
- Le réseau 192.168.1.16/30 est atteint également via la passerelle 192.168.1.0 avec une métrique de 120.
- Le réseau 192.168.1.20/30 est directement connecté via l'interface Ethernet1/0.

- Enfin, il y a une route locale (L) pour chaque adresse IP des interfaces FastEthernet0/0 et Ethernet1/0 (192.168.1.1 et 192.168.1.21 respectivement).

❖ SUPPRESSION DE LA LIAISON ENTRE R2 ET R3 ET DESACTIVATION DES INTERFACES CORRESPONDANTES :



• Temps de jonction des routeurs :

Dans RIP, le routeur R1 peut joindre R3 dans un temps typique jusqu'à 180 secondes.

```
R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 140/178/228 ms
R1#
```

❖ ENCAPSULATIONS DES PAQUETS RIP

Avec le protocole de routage RIP v2, le temps nécessaire à R1 pour détecter les changements dans la topologie du réseau et mettre à jour sa table de routage dépendra des paramètres de temporisation RIP v2 configurés sur les routeurs.

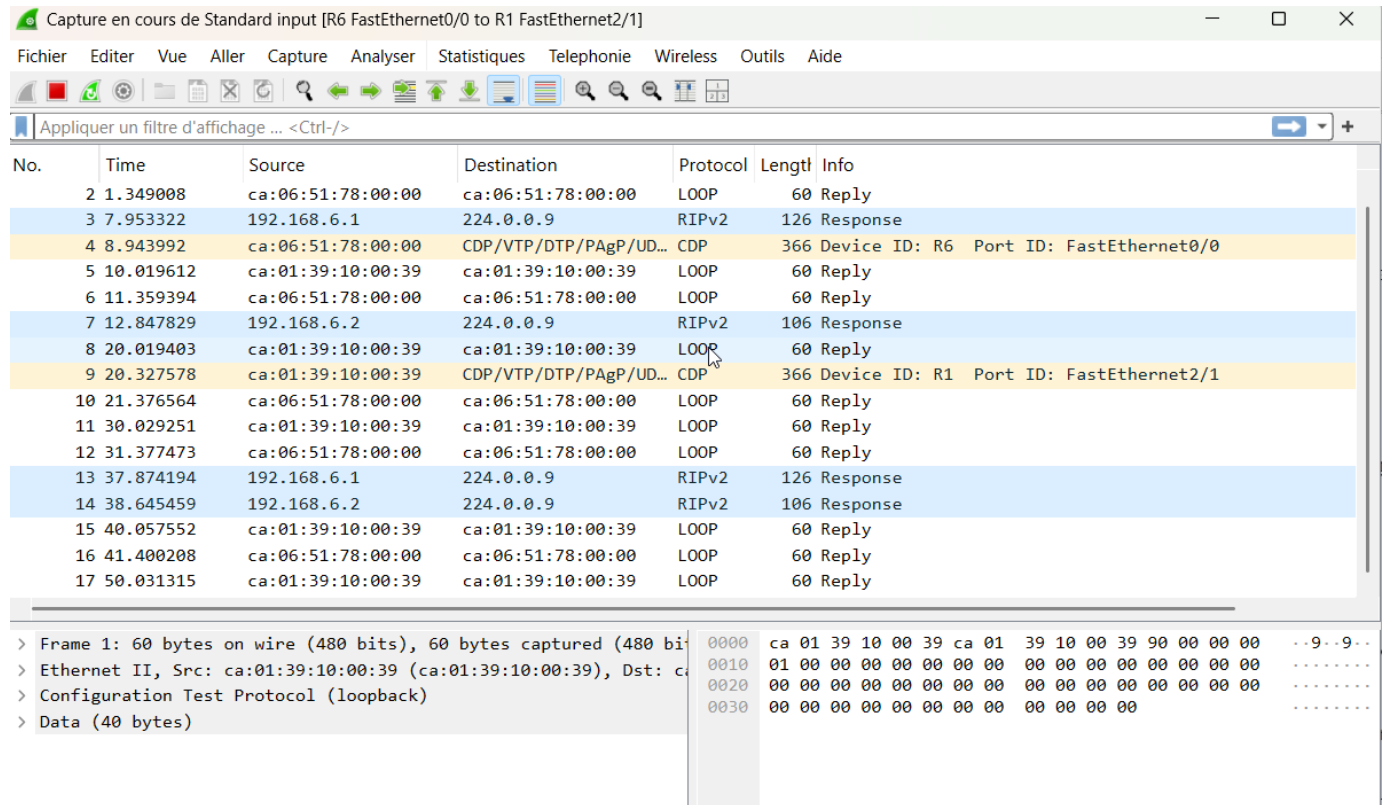
Par défaut, RIP v2 utilise une temporisation de mise à jour de 30 secondes, ce qui signifie que R1 enverra des mises à jour de sa table de routage à ses voisins RIP toutes les 30 secondes. Après la suppression du lien entre R2 et R3 et la désactivation des interfaces correspondantes sur le réseau 192.168.2.0/30, R1 devra attendre 120s pour atteindre le router R3.

Une fois que R1 détecte le changement, il commencera le processus de convergence RIP, qui peut prendre plusieurs mises à jour du protocole avant que la nouvelle topologie soit pleinement reflétée dans sa table de routage.

• Numéro de Port :

Les paquets RIP sont encapsulé dans le Protocole UDP, numero de port 520.

❖ REMISE DE LIEN ENTRE R2 ET R3 :



No.	Time	Source	Destination	Protocol	Length	Info
2	1.349008	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
3	7.953322	192.168.6.1	224.0.0.9	RIPv2	126	Response
4	8.943992	ca:06:51:78:00:00	CDP/VTP/DTP/PagP/UD...	CDP	366	Device ID: R6 Port ID: FastEthernet0/0
5	10.019612	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
6	11.359394	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
7	12.847829	192.168.6.2	224.0.0.9	RIPv2	106	Response
8	20.019403	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
9	20.327578	ca:01:39:10:00:39	CDP/VTP/DTP/PagP/UD...	CDP	366	Device ID: R1 Port ID: FastEthernet2/1
10	21.376564	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
11	30.029251	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
12	31.377473	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
13	37.874194	192.168.6.1	224.0.0.9	RIPv2	126	Response
14	38.645459	192.168.6.2	224.0.0.9	RIPv2	106	Response
15	40.057552	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
16	41.400208	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
17	50.031315	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
Ethernet II, Src: ca:01:39:10:00:39 (ca:01:39:10:00:39), Dst: ca:01:39:10:00:39 (ca:01:39:10:00:39)
Configuration Test Protocol (loopback)
Data (40 bytes)

• Observation :

- Il existe des nouveaux paquets envoyés entre le routeur R2 et R3.
- Il y a un changement du comportement des réseaux.

• Commentaires :

Request/Response : Ces messages sont utilisés pour demander ou répondre aux mises à jour périodiques de la table de routage. Chaque routeur envoie périodiquement un message Request pour demander des mises à jour de ses voisins (avec l'adresse multicast 224.0.0.9 et le port UDP 520). Les voisins répondent ensuite avec des messages Response contenant les informations de leur table de routage.

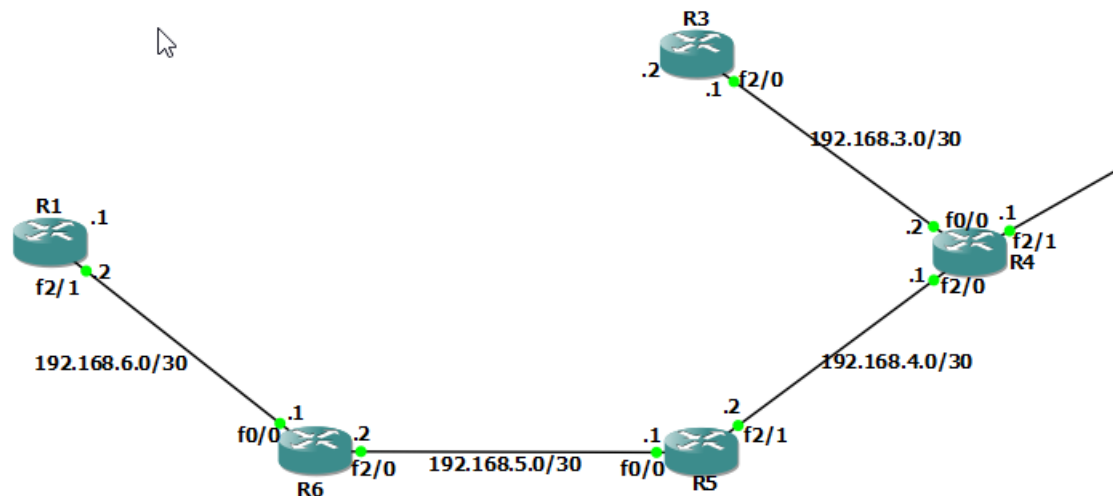
Il y a donc des échanges de message RIPv2 entre les routeurs en utilisant l'adresse multicast ci-dessus.

En cas de rupture de liaison, les messages disparaîtraient et entraînant une convergence du réseau et suppression des routes invalides.

❖ SUPPRESSION DU ROUTEUR R2

Entrainant la perte des échanges de mises à jour de routage ;

Présentation de la suppression :



Routeur R2 supprimé, ce qui entraîne la suppression des réseaux 192.168.1.0 et 192.168.1.2, cela déclenchera une convergence du protocole de routage RIP sur les routeurs R1 et R3.

• Messages échangés :

Dans Wireshark,

Capture en cours de Standard input [R6 FastEthernet0/0 to R1 FastEthernet2/1]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
11	41.533200	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
12	42.492580	ca:01:39:10:00:39	ca:01:39:10:00:39	CDP/VTP/DTP/PagP/UD...	366	Device ID: R1 Port ID: FastEthernet2/1
13	49.498770	ca:06:51:78:00:00	ca:06:51:78:00:00	CDP/VTP/DTP/PagP/UD...	366	Device ID: R6 Port ID: FastEthernet0/0
14	50.040322	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
15	51.549278	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
16	53.118125	192.168.6.1	224.0.0.9	RIPv2	246	Response
17	60.045733	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
18	61.560206	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
19	70.046027	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
20	71.573954	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
21	79.158276	192.168.6.1	224.0.0.9	RIPv2	246	Response
22	80.046052	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
23	81.573608	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply
24	90.059755	ca:01:39:10:00:39	ca:01:39:10:00:39	LOOP	60	Reply
25	91.249636	ca:01:39:10:00:39	ca:01:39:10:00:39	CDP/VTP/DTP/PagP/UD...	366	Device ID: R1 Port ID: FastEthernet2/1
26	91.572919	ca:06:51:78:00:00	ca:06:51:78:00:00	LOOP	60	Reply

> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
> Ethernet II, Src: ca:01:39:10:00:39 (ca:01:39:10:00:39), Dst: ca:01:39:10:00:39 (ca:01:39:10:00:39)
> Configuration Test Protocol (loopback)
> Data (40 bytes)

Sortie de R1 :

-R1 enverra des messages RIP Response pour informer ses voisins de la suppression des routes vers les réseaux 192.168.1.0 et 192.168.1.2.

-R1 recevra également des messages RIP Update des autres routeurs, probablement R6, pour indiquer la même chose.

Capture en cours de Standard input [R3 FastEthernet2/0 to R4 FastEthernet0/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.3.2	224.0.0.9	RIPv2	246	Response
2	4.306700	ca:04:03:48:00:00	ca:04:03:48:00:00	LOOP	60	Reply
3	4.356566	ca:03:46:20:00:38	ca:03:46:20:00:38	LOOP	60	Reply
4	14.304582	ca:04:03:48:00:00	ca:04:03:48:00:00	LOOP	60	Reply
5	14.337494	ca:03:46:20:00:38	ca:03:46:20:00:38	LOOP	60	Reply
6	24.315095	ca:04:03:48:00:00	ca:04:03:48:00:00	LOOP	60	Reply
7	24.349004	ca:03:46:20:00:38	ca:03:46:20:00:38	LOOP	60	Reply
8	27.464855	192.168.3.2	224.0.0.9	RIPv2	246	Response
9	34.327134	ca:04:03:48:00:00	ca:04:03:48:00:00	LOOP	60	Reply
10	34.368144	ca:03:46:20:00:38	ca:03:46:20:00:38	LOOP	60	Reply
11	34.614392	ca:03:46:20:00:38	CDP/VTP/DTP/PAGP/UD...	CDP	366	Device ID: R3 Port ID: FastEthernet2/0
12	44.331879	ca:04:03:48:00:00	ca:04:03:48:00:00	LOOP	60	Reply
13	44.359804	ca:03:46:20:00:38	ca:03:46:20:00:38	LOOP	60	Reply
14	46.625756	ca:04:03:48:00:00	CDP/VTP/DTP/PAGP/UD...	CDP	366	Device ID: R4 Port ID: FastEthernet0/0
15	54.242524	192.168.3.2	224.0.0.9	RIPv2	246	Response
16	54.316325	ca:04:03:48:00:00	ca:04:03:48:00:00	LOOP	60	Reply
17	54.352229	ca:03:46:20:00:38	ca:03:46:20:00:38	LOOP	60	Reply
18	64.317573	ca:04:03:48:00:00	ca:04:03:48:00:00	LOOP	60	Reply
19	64.354474	ca:03:46:20:00:38	ca:03:46:20:00:38	LOOP	60	Reply

Standard input: <live capture in progress> Paquets : 19 Profil : Default

Frame 1: 246 bytes on wire (1968 bits), 246 bytes captured (1968 bits) on Standard input
 Ethernet II, Src: ca:04:03:48:00:00 (ca:04:03:48:00:00), Dst: 01:00:5e:00:00:09
 Internet Protocol Version 4, Src: 192.168.3.2, Dst: 224.0.0.9
 User Datagram Protocol, Src Port: 520, Dst Port: 520
 Routing Information Protocol

0000 01 00 5e 00 00 09 ca 04 03 48 00 00 08 00 45 c0 ..^...
 0010 00 e8 00 00 00 00 02 11 13 92 c0 a8 03 02 e0 00
 0020 00 09 02 08 02 08 00 d4 13 03 02 02 00 00 00 02
 0030 00 00 0a 00 00 07 ff ff ff ff 00 00 00 00 00 00
 0040 00 05 00 02 00 00 0a 00 00 08 ff ff ff ff 00 00
 0050 00 00 00 00 00 05 00 02 00 00 0a 00 00 09 ff ff
 0060 ff ff 00 00 00 00 00 00 00 05 00 02 00 00 ac 10
 0070 07 00 ff ff ff ff 00 00 00 00 00 00 00 00 00 00

Sortie de R3 :

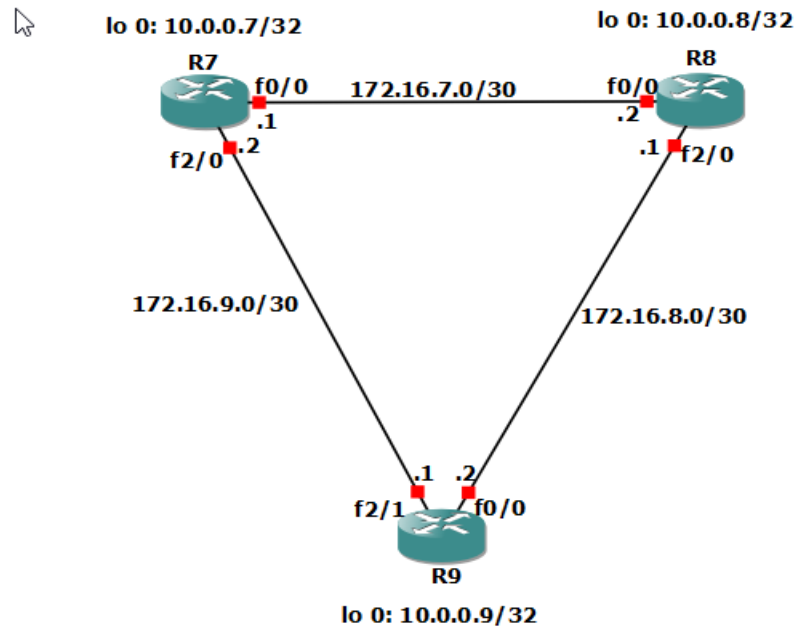
- R3 enverra des messages RIP Response pour informer ses voisins de la suppression des routes vers les réseaux 192.168.1.0 et 192.168.1.2.
- R3 recevra également des messages RIP Update des autres routeurs, probablement R4, pour indiquer la même chose.

• Temps de convergence :

La convergence peut prendre vers 180 secondes après la considération que le routeur R2 est en panne.

Partie OSPF et Redistribution

➤ Voici la topologie OSPF :



❖ CONFIGURATION DES ROUTEURS ET ACTIVATION DES INTERFACES DU RESEAU

Configuration des routeurs de la partie OSPF, on va voir toutes les configurations des routeurs (R7, R8 et R9) ci-après avec l'activation des interfaces.

CONFIGURATION DU ROUTEUR R7 :

```
R1 R2 R3 R4 R5 R6 R7 x R8 R9 | + - □ ×

R7#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#int lo 0
R7(config-if)#
*Apr 30 09:28:21.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R7(config-if)#ip add 10.0.0.7 255.255.255.255
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#int f2/1
R7(config-if)#ip add 172.16.10.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:30:34.367: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:30:35.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R7(config)#int f0/0
R7(config-if)#ip add 172.16.7.1 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
*Apr 30 09:31:32.599: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:31:33.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R7(config-if)#exit
R7(config)#int f2/0
R7(config-if)#ip add 172.16.9.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:32:50.631: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
R7(config)#
*Apr 30 09:32:51.631: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0

R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0
R7(config-router)#
*Apr 30 09:35:13.879: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet2/1 from LOADING to FULL, Loading Done
R7(config-router)#exit
R7(config)#redistribute rip subnets metric 20 metric-type 2
^
% Invalid input detected at '^' marker.

R7(config)#redistribute rip subnets metric 20 metric-type 2
^
% Invalid input detected at '^' marker.

R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0
R7(config-router)#redistribute rip subnets metric 20 metric-type 2
R7(config-router)#exit
R7(config)#router rip
```

CONFIGURATION DU ROUTEUR R8 :

```
R1 R2 R3 R4 R5 R6 R7 R8 R9
R8#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R8(config)#int lo 0
R8(config-if)#
*Apr 30 09:44:01.859: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R8(config-if)#ip add 10.0.0.8 255.255.255.255
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#int f0/0
R8(config-if)#ip add 172.16.7.2 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
R8(config)#
*Apr 30 09:45:54.059: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:45:55.059: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R8(config)#int f2/0
R8(config-if)#ip add 172.16.8.1 255.255.255.252
R8(config-if)#no shutdown
R8(config-if)#exit
*Apr 30 09:46:38.195: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:46:39.195: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R8(config-if)#exit
R8(config)#router ospf 1
R8(config-router)#router-id 8.8.8.8
R8(config-router)#network 10.0.0.8 0.0.0.0 area 0
R8(config-router)#network 172.16.7.0 0.0.0.3 area 0
R8(config-router)#network 172.16.8.0 0.0.0.3 area 0
*Apr 30 09:48:37.539: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on FastEthernet0/0 from LOADING to FULL, Loading Done
R8(config-router)#network 172.16.8.0 0.0.0.3 area 0
R8(config-router)#
*Apr 30 09:54:36.267: %OSPF-5-ADJCHG: Process 1, Nbr 9.9.9.9 on FastEthernet2/0 from LOADING to FULL, Loading Done
R8(config-router)#end
R8#w
*Apr 30 09:58:40.007: %SYS-5-CONFIG_I: Configured from console by console
R8#wr
Warning: Attempting to overwrite an NVRAM configuration previously written
```

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CONFIGURATION DU ROUTEUR R9 :

```
R1 R2 R3 R4 R5 R6 R7 R8 R9 x + - □ ×
R9#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R9(config)#int lo 0
R9(config-if)#i
*Apr 30 09:50:31.759: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R9(config-if)#ip add 10.0.0.9 255.255.255.255
R9(config-if)#no shutdown
R9(config-if)#exit
R9(config)#int f0/0
R9(config-if)#ip add 172.16.8.2 255.255.255.252
R9(config-if)#no shutdown
R9(config-if)#exit
R9(config)#
*Apr 30 09:51:52.275: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:51:53.279: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R9(config)#int f2/1
R9(config-if)#ip add 172.16.9.1 255.255.255.252
R9(config-if)#no shutdown
R9(config-if)#exit
R9(config)#
*Apr 30 09:52:38.771: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:52:39.771: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R9(config)#router ospf 1
R9(config-router)#router-id 9.9.9.9
R9(config-router)#network 10.0.0.9 0.0.0.0 area 0
R9(config-router)#network 172.16.9.0 0.0.0.3 area 0
R9(config-router)#network 172.16.9.0 0.0.0.3 area 0
*Apr 30 09:53:58.443: %OSPF-5-ADJCHG: Process 1, Nbr 7.7.7.7 on FastEthernet2/1 from LOADING to FULL, Loading Done
R9(config-router)#network 172.16.8.0 0.0.0.3 area 0
R9(config-router)#
*Apr 30 09:54:36.451: %OSPF-5-ADJCHG: Process 1, Nbr 8.8.8.8 on FastEthernet0/0 from LOADING to FULL, Loading Done
R9(config-router)#end
R9#
*Apr 30 09:58:57.867: %SYS-5-CONFIG_I: Configured from console by console
R9#wr
Warning: Attempting to overwrite an NVRAM configuration previously written

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```

Toutes les configurations des routeurs dans l'OSPF sont tous faites.

❖ MESSAGES ECHANGES ENTRE LES ROUTEURS

Capture en cours de Standard input [R9 FastEthernet2/1 to R7 FastEthernet2/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ca:07:0f:a4:00:38	ca:07:0f:a4:00:38	LOOP	60	Reply
2	2.393855	172.16.9.2	224.0.0.5	OSPF	94	Hello Packet
3	3.400424	ca:07:0f:a4:00:38	CDP/VTP/DTP/PagP/UD...	CDP	366	Device ID: R7 Port ID: FastEthernet2/0
4	6.106754	172.16.9.1	224.0.0.5	OSPF	94	Hello Packet
5	6.658488	ca:09:3d:48:00:39	ca:09:3d:48:00:39	LOOP	60	Reply
6	9.972579	ca:07:0f:a4:00:38	ca:07:0f:a4:00:38	LOOP	60	Reply
7	9.994862	172.16.9.2	224.0.0.9	RIPv2	166	Response
8	11.711785	172.16.9.2	224.0.0.5	OSPF	94	Hello Packet
9	15.432181	172.16.9.1	224.0.0.5	OSPF	94	Hello Packet
10	16.656885	ca:09:3d:48:00:39	ca:09:3d:48:00:39	LOOP	60	Reply
11	19.964585	ca:07:0f:a4:00:38	ca:07:0f:a4:00:38	LOOP	60	Reply
12	20.884950	172.16.9.2	224.0.0.5	OSPF	94	Hello Packet
13	25.067743	172.16.9.1	224.0.0.5	OSPF	94	Hello Packet
14	26.648888	ca:09:3d:48:00:39	ca:09:3d:48:00:39	LOOP	60	Reply
15	29.966192	ca:07:0f:a4:00:38	ca:07:0f:a4:00:38	LOOP	60	Reply
16	30.681623	172.16.9.2	224.0.0.5	OSPF	94	Hello Packet

> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

> Ethernet II, Src: ca:07:0f:a4:00:38 (ca:07:0f:a4:00:38), Dst: ca:07:0f:a4:00:38

> Configuration Test Protocol (loopback)

> Data (40 bytes)

```

0000  ca 07 0f a4 00 38 ca 07 0f a4 00 38 90 00 00 00  ....8..
0010  01 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ....
0020  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ....
0030  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ....

```

❖ PAQUET HELLO D'OSPF

[R7 Ethernet1/0 to R9 Ethernet1/0]

Fichier Editer Vue Aller Capture Analyser Statistiques Telephonie Wireless Outils Aide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ca:07:13:f4:00:1c	CDP/VTP/DTP/PagP/UD...	CDP	362	Device ID: R7 Port ID: Ethernet1/0
2	1.000159	172.16.1.6	224.0.0.5	OSPF	94	Hello Packet
3	3.000144	ca:07:13:f4:00:1c	ca:07:13:f4:00:1c	LOOP	60	Reply
4	6.420802	172.16.1.5	224.0.0.5	OSPF	94	Hello Packet
5	7.155004	ca:09:2b:e0:00:1c	ca:09:2b:e0:00:1c	LOOP	60	Reply
6	10.826015	172.16.1.6	224.0.0.5	OSPF	94	Hello Packet
7	13.903380	ca:07:13:f4:00:1c	ca:07:13:f4:00:1c	LOOP	60	Reply
8	15.574905	172.16.1.5	224.0.0.5	OSPF	94	Hello Packet
9	17.155553	ca:09:2b:e0:00:1c	ca:09:2b:e0:00:1c	LOOP	60	Reply
10	17.824335	ca:09:2b:e0:00:1c	CDP/VTP/DTP/PagP/UD...	CDP	362	Device ID: R9 Port ID: Ethernet1/0
11	20.261263	172.16.1.6	224.0.0.5	OSPF	94	Hello Packet
12	23.917086	ca:07:13:f4:00:1c	ca:07:13:f4:00:1c	LOOP	60	Reply
13	25.026206	172.16.1.5	224.0.0.5	OSPF	94	Hello Packet
14	27.155664	ca:09:2b:e0:00:1c	ca:09:2b:e0:00:1c	LOOP	60	Reply
15	30.212482	172.16.1.6	224.0.0.5	OSPF	94	Hello Packet

> Frame 6: 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface 0, id 0

> Ethernet II, Src: ca:09:2b:e0:00:1c (ca:09:2b:e0:00:1c), Dst: IPv4multicast_05 (01:00:5e:00:00:05)

> Internet Protocol Version 4, Src: 172.16.1.6, Dst: 224.0.0.5

> Open Shortest Path First

> OSPF Header

> OSPF Hello Packet

Network Mask: 255.255.255.252

Hello Interval [sec]: 10

Options: 0x12, (L) LLS Data Block, (E) External Routing

Router Priority: 1

Router Dead Interval [sec]: 40

Designated Router: 172.16.1.5

Backup Designated Router: 172.16.1.6

Active Neighbor: 10.0.0.7

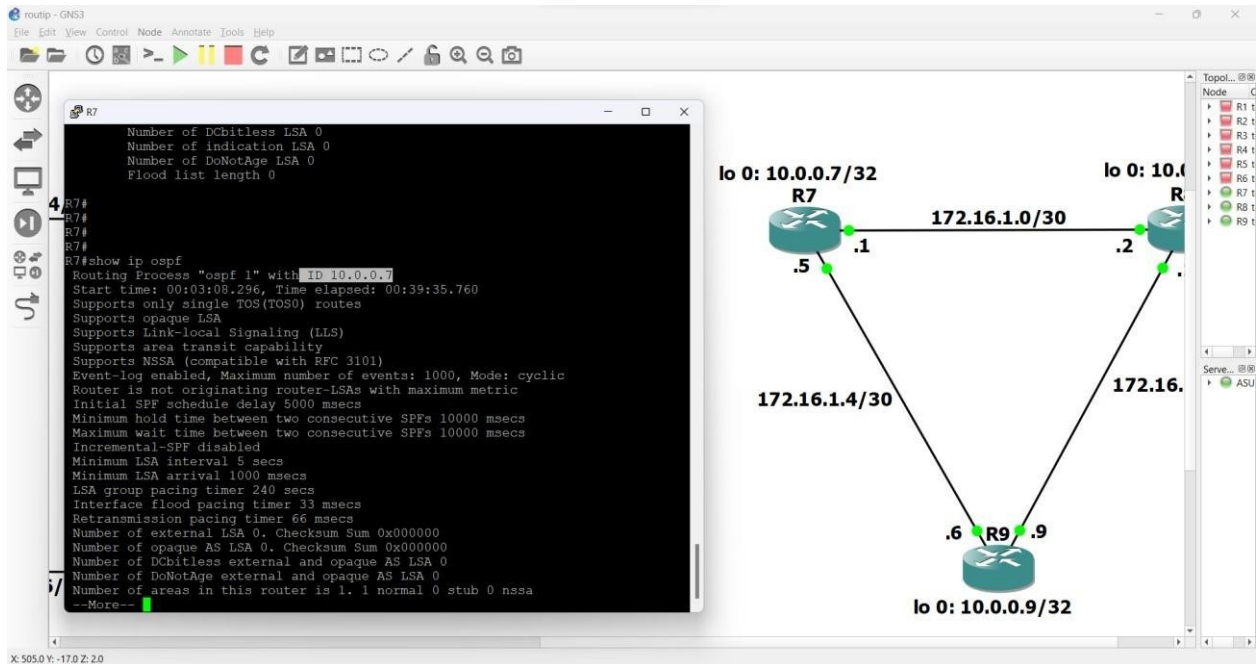
> OSPF LLS Data Block

Paquets : 15 - Affichés : 15 (100.0%) - Perdus : 0 (0.0%) - Profil : Default

X.000.Y.000.Z.0.5

On peut voir la durée théorique entre deux messages Hello d'OSPF est 10 secondes (Hello interval).

❖ LES IDENTIFIANTS PRIS PAR CHAQUE ROUTEUR



Si un identifiant de routeur n'est pas explicitement configuré, OSPF choisira l'une des adresses IP de ses interfaces comme identifiant de routeur. Il préfère utiliser l'adresse IP de l'interface loopback si elle est configurée, ici il prend l'adresse 10.0.0.7 comme ID de routeur R7.

❖ TEST DE CONNECTIVITE (PING)

- **ROUTEUR R7 vers les autres routeurs dans OSPF :**
Connectivité aux routeurs


```
by a different version of the system image.
Overwrite the previous NVRAM configuration?[confirm]
Building configuration...
[OK]
R7#ping 192.168.6.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 148/177/212 ms
R7#
*Apr 30 11:09:17.495: %OSPF-5-ADJCHG: Process 1, Nbr 4.4.4.4 on FastEthernet2/1 from FULL to DOWN, Neighbor Down: Dead timer expired
R7#ping 172.16.7.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/5/8 ms
R7#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 36/40/48 ms
R7#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/42/68 ms
R7#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/67/112 ms
R7#ping 172.16.9.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/28/44 ms
R7#ping 172.16.9.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/8 ms
R7#
```

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• ROUTEUR R8 :

Ping vers 172.16.8.2

Ping vers 172.16.9.1

Ping vers 172.16.9.2

Ping vers 172.16.7.1

Ping vers 172.16.7.2

```

R8#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/46/60 ms
R8#ping 172.16.9.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/72/88 ms
R8#ping 172.16.9.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/40/52 ms
R8#ping 172.16.7.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 28/37/44 ms
R8#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R8#

```

- ROUTEUR R9 :

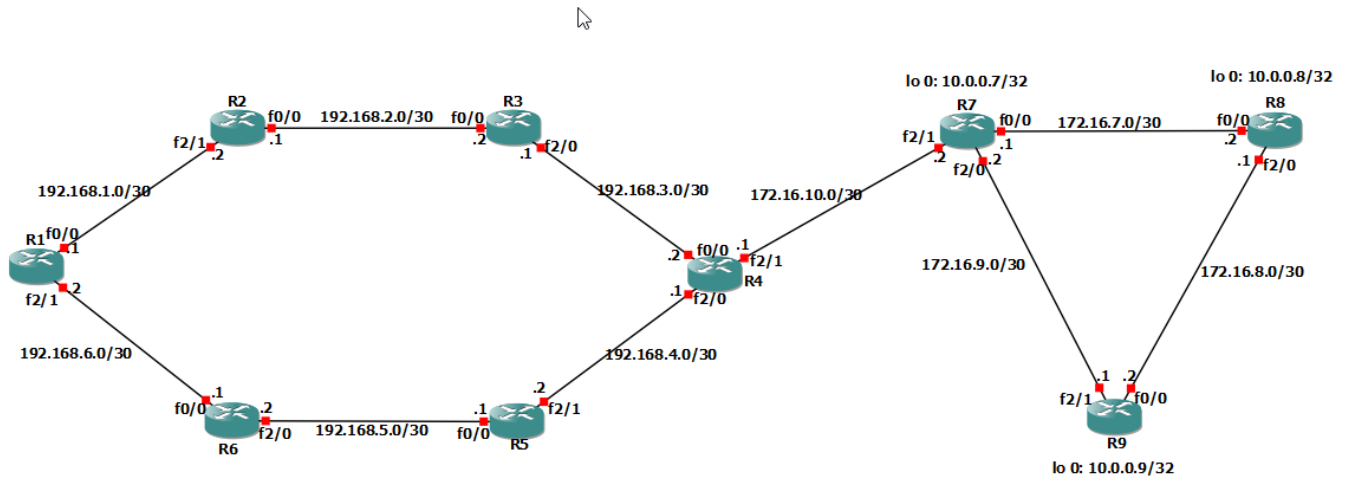
```

[OK]
R9#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/8/12 ms
R9#ping 172.16.9.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/48/80 ms
R9#ping 172.16.7.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 44/71/88 ms
R9#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/55/84 ms
R9#ping 172.16.8.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/3/8 ms
R9#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/56/76 ms
R9#ping 172.16.9.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.9.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
R9#

```

❖ REDISTRIBUTION DES ROUTES ENTRE LES PROTOCOLES RIP et OSPF

- Liaison de RIP et OSPF via R4 et R7 :



- Vérification des tables de routage dans le réseau RIP

Les tables de routage doivent être bien assurées pour que la configuration soit correcte.

```
10.0.0.0/32 is subnetted, 3 subnets
R    10.0.0.7 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    10.0.0.8 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    10.0.0.9 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
172.16.0.0/30 is subnetted, 4 subnets
R    172.16.7.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    172.16.8.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    172.16.9.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R    172.16.10.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, FastEthernet0/0
L    192.168.1.1/32 is directly connected, FastEthernet0/0
192.168.2.0/30 is subnetted, 1 subnets
R    192.168.2.0 [120/1] via 192.168.1.2, 00:00:24, FastEthernet0/0
192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/2] via 192.168.1.2, 00:00:24, FastEthernet0/0
192.168.4.0/30 is subnetted, 1 subnets
R    192.168.4.0 [120/2] via 192.168.6.1, 00:00:04, FastEthernet2/1
192.168.5.0/30 is subnetted, 1 subnets
R    192.168.5.0 [120/1] via 192.168.6.1, 00:00:04, FastEthernet2/1
192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.6.0/30 is directly connected, FastEthernet2/1
L    192.168.6.2/32 is directly connected, FastEthernet2/1
R1#
R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 140/178/228 ms
R1#
```

❖ ACTIVATION DES NOUVELLES INTERFACES ET ATTRIBUTION D'UNE ADRESSE IP / MASQUE

Ici on va attribuer des adresse IP entre R4 et R7, alors on va configurer et activer OSPF sur ces deux nouvelles interfaces.

- ROUTEUR R4 :



```
*Apr 30 09:12:20.243: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to down
R4#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R4(config)#int f0/0
R4(config-if)#ip add 192.168.3.2 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:08.539: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:16:09.539: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R4(config)#int f2/0
R4(config-if)#ip add 192.168.4.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:16:58.587: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
*Apr 30 09:16:59.587: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R4(config)#int f2/1
R4(config-if)#ip add 172.16.10.1 255.255.255.252
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
*Apr 30 09:17:49.963: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:17:50.963: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
R4(config-router)#
```

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- ROUTEUR R7 :

```
R1 R2 R3 R4 R5 R6 R7 x R8 R9 + - □ ×
R7#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R7(config)#int lo 0
R7(config-if)#
*Apr 30 09:28:21.827: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up
R7(config-if)#ip add 10.0.0.7 255.255.255.255
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#int f2/1
R7(config-if)#ip add 172.16.10.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:30:34.367: %LINK-3-UPDOWN: Interface FastEthernet2/1, changed state to up
*Apr 30 09:30:35.367: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/1, changed state to up
R7(config)#int f0/0
R7(config-if)#ip add 172.16.7.1 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
*Apr 30 09:31:32.599: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Apr 30 09:31:33.599: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R7(config-if)#exit
R7(config)#int f2/0
R7(config-if)#ip add 172.16.9.2 255.255.255.252
R7(config-if)#no shutdown
R7(config-if)#exit
R7(config)#
*Apr 30 09:32:50.631: %LINK-3-UPDOWN: Interface FastEthernet2/0, changed state to up
R7(config)#
*Apr 30 09:32:51.631: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet2/0, changed state to up
R7(config)#router ospf 1
R7(config-router)#router-id 7.7.7.7
R7(config-router)#network 10.0.0.7 0.0.0.0 area 0
R7(config-router)#network 172.16.7.0 0.0.0.3 area 0
R7(config-router)#network 172.16.9.0 0.0.0.3 area 0
R7(config-router)#network 172.16.10.0 0.0.0.3 area 0

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```

Après cette configuration et activation des nouvelles interfaces, on va voir si R4 peut ping l'interface loopback de R8.

❖ PING R4 SUR R8

```
R4#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/108/156 ms
R4#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/68/80 ms
R4#ping 10.0.0.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/73/112 ms
R4#
```

On voit que R4 peut ping l'interface loopback de R8 car R4 est connecté directement au réseau OSPF via R7 dont R8 est dans le domaine OSPF et aussi parce que les loopback sont déclarées dans OSPF, entraînant les échanges automatiques des routes.

❖ PING DE R8 SUR L'INTERFACE R4 CONNECTE AU ROUTEUR R3

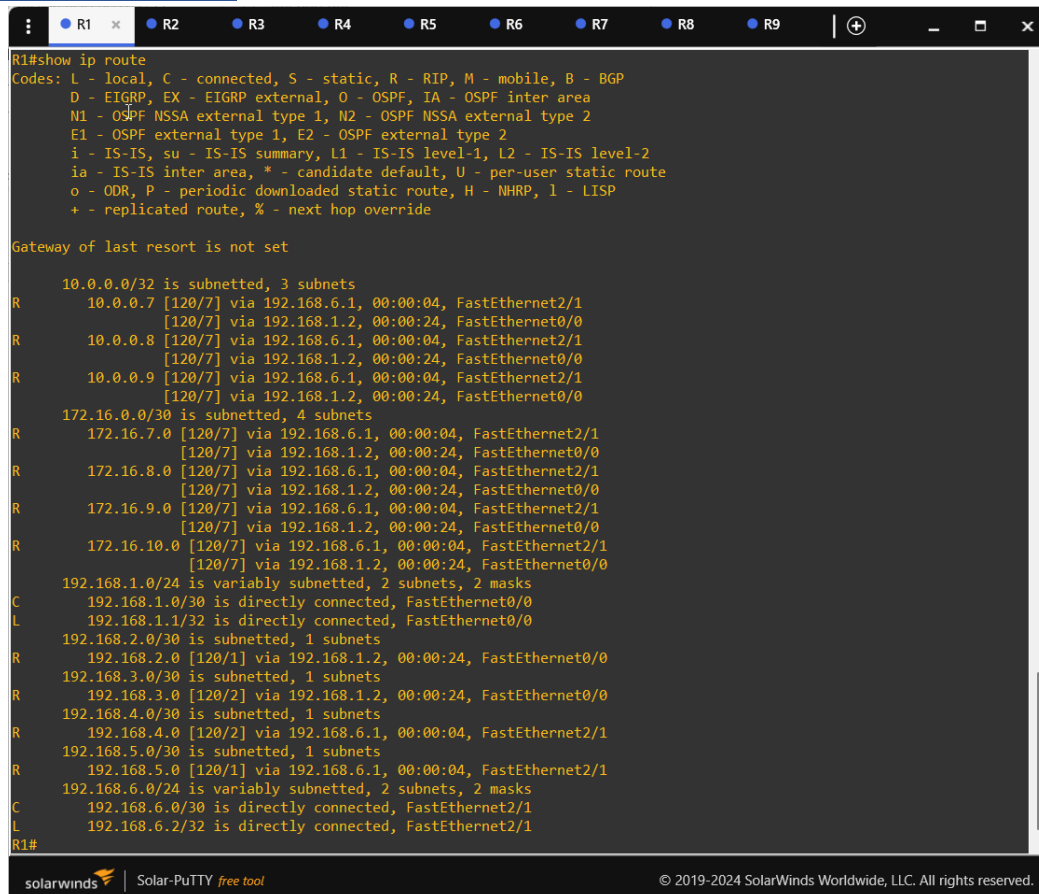
On peut constater que le routeur R8 ne peut pas ping l'interface de R4 qui est connecté à R3 car les routeurs OSPF n'ont pas configuré le chemin vers les réseaux RIP et aussi que R8 est dans le domaine OSPF tandis que R4 et R3 appartiennent au domaine RIP. De plus, à notre stade là, aucune redistribution n'est encore faite.

❖ PING ENTRE R1 ET R8

Dans ce cas, R1 ne peut pas ping l'interface de R8 et aussi inversement parce qu'il n'y a pas encore de redistribution entre RIP et OSPF alors qu'aucun chemin valide n'est trouvé entre eux par les routeurs.

❖ TABLE DE ROUTAGE DES ROUTEURS :

• ROUTEUR R1 :



```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       * - replicated route, % - next hop override

Gateway of last resort is not set

 10.0.0.0/32 is subnetted, 3 subnets
R   10.0.0.7 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R   10.0.0.8 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R   10.0.0.9 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
 172.16.0.0/30 is subnetted, 4 subnets
R   172.16.7.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R   172.16.8.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R   172.16.9.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
R   172.16.10.0 [120/7] via 192.168.6.1, 00:00:04, FastEthernet2/1
    [120/7] via 192.168.1.2, 00:00:24, FastEthernet0/0
 192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.1.0/30 is directly connected, FastEthernet0/0
L   192.168.1.1/32 is directly connected, FastEthernet0/0
 192.168.2.0/30 is subnetted, 1 subnets
R   192.168.2.0 [120/1] via 192.168.1.2, 00:00:24, FastEthernet0/0
 192.168.3.0/30 is subnetted, 1 subnets
R   192.168.3.0 [120/2] via 192.168.1.2, 00:00:24, FastEthernet0/0
 192.168.4.0/30 is subnetted, 1 subnets
R   192.168.4.0 [120/2] via 192.168.6.1, 00:00:04, FastEthernet2/1
 192.168.5.0/30 is subnetted, 1 subnets
R   192.168.5.0 [120/1] via 192.168.6.1, 00:00:04, FastEthernet2/1
 192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.6.0/30 is directly connected, FastEthernet2/1
L   192.168.6.2/32 is directly connected, FastEthernet2/1
R1#
```

- ROUTEUR R4 :

```
R4#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/32 is subnetted, 3 subnets
O       10.0.0.7 [110/2] via 172.16.10.2, 00:06:26, FastEthernet2/1
O       10.0.0.8 [110/3] via 172.16.10.2, 00:06:26, FastEthernet2/1
O       10.0.0.9 [110/3] via 172.16.10.2, 00:06:26, FastEthernet2/1
    172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
O       172.16.7.0/30 [110/2] via 172.16.10.2, 00:06:26, FastEthernet2/1
O       172.16.8.0/30 [110/3] via 172.16.10.2, 00:06:26, FastEthernet2/1
O       172.16.9.0/30 [110/2] via 172.16.10.2, 00:06:26, FastEthernet2/1
C       172.16.10.0/30 is directly connected, FastEthernet2/1
L       172.16.10.1/32 is directly connected, FastEthernet2/1
    192.168.1.0/30 is subnetted, 1 subnets
R       192.168.1.0 [120/2] via 192.168.3.1, 00:00:17, FastEthernet0/0
    192.168.2.0/30 is subnetted, 1 subnets
R       192.168.2.0 [120/1] via 192.168.3.1, 00:00:17, FastEthernet0/0
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/30 is directly connected, FastEthernet0/0
L       192.168.3.2/32 is directly connected, FastEthernet0/0
    192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.4.0/30 is directly connected, FastEthernet2/0
L       192.168.4.1/32 is directly connected, FastEthernet2/0
    192.168.5.0/30 is subnetted, 1 subnets
R       192.168.5.0 [120/1] via 192.168.4.2, 00:00:10, FastEthernet2/0
    192.168.6.0/30 is subnetted, 1 subnets
R       192.168.6.0 [120/2] via 192.168.4.2, 00:00:10, FastEthernet2/0
R4#
```

- ROUTEUR R8 :

❖ REDISTRIBUTION DES ROUTES DE RIP SUR OSPF AU NIVEAU DE R4

Maintenant on va redistribuer les routes de RIP sur OFSP au niveau de R4 :


```

R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2

```

Sur le routeur R4, une redistribution est mise en œuvre afin d'intégrer les routes du protocole RIP dans le domaine OSPF, permettant ainsi d'assurer la communication et l'échange d'informations de routage entre ces deux protocoles distincts.

Après la redistribution de route des RIP sur OSPF, une Route vers R1 apparait dans la table de routage de routeur R8 :

✓ TABLE DE ROUTAGE DU ROUTEUR R8 :

```

R8#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       I  N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
          E1 - OSPF external type 1, E2 - OSPF external type 2
          i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
          ia - IS-IS inter area, * - candidate default, U - per-user static route
          o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
          + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/32 is subnetted, 3 subnets
O       10.0.0.7 [110/2] via 172.16.7.1, 02:03:00, FastEthernet0/0
C       10.0.0.8 is directly connected, Loopback0
O       10.0.0.9 [110/2] via 172.16.8.2, 01:57:01, FastEthernet2/0
    172.16.0.0/16 is variably subnetted, 6 subnets, 2 masks
C       172.16.7.0/30 is directly connected, FastEthernet0/0
L       172.16.7.2/32 is directly connected, FastEthernet0/0
C       172.16.8.0/30 is directly connected, FastEthernet2/0
L       172.16.8.1/32 is directly connected, FastEthernet2/0
O       172.16.9.0/30 [110/2] via 172.16.8.2, 01:57:01, FastEthernet2/0
          [110/2] via 172.16.7.1, 02:03:00, FastEthernet0/0
O       172.16.10.0/30 [110/2] via 172.16.7.1, 00:42:25, FastEthernet0/0
    192.168.1.0/30 is subnetted, 1 subnets
O E2    192.168.1.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
    192.168.2.0/30 is subnetted, 1 subnets
O E2    192.168.2.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
    192.168.3.0/30 is subnetted, 1 subnets
O E2    192.168.3.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
    192.168.4.0/30 is subnetted, 1 subnets
O E2    192.168.4.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
    192.168.5.0/30 is subnetted, 1 subnets
O E2    192.168.5.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
    192.168.6.0/30 is subnetted, 1 subnets
O E2    192.168.6.0 [110/20] via 172.16.7.1, 00:08:20, FastEthernet0/0
R8#

```

Alors on va effectuer un ping de R8 vers une interface de R1 ;

❖ FONCTIONNEMENT DU PING DE R8 Vers UNE INTERFACE DE R1 :

```
R8#ping 192.168.6.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 148/156/164 ms
R8#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 128/149/168 ms
R8#
```

D'après ce qu'on voit, on peut déterminer que le ping de R8 vers une interface de R1 fonctionne très bien.

• Paquets ICMP :

Le paquet ICMP arrive à R1 mais R1 ne peut pas envoyer des réponses car le chemin vers les réseaux OSPF n'existe pas dans la table de routage de R1 (retour non garanti).

❖ PING DE R1 VERS R8 :

Le routeur R1 peut maintenant pinger le routeur R8 parce que la redistribution est déjà faite entre RIP et OSPF.

❖ REDISTRIBUTION DES TABLES DE ROUTAGE OSPF SUR RIP AU NIVEAU DE R4

Ici, on va redistribuer les tables de routage OSPF sur RIP au niveau de R4.

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
```

Et puis regardons l'une des tables de routage des routeurs RIP, des modifications doivent y parvenir :

```

Gateway of last resort is not set

10.0.0.0/32 is subnetted, 3 subnets
R    10.0.0.7 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    10.0.0.8 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    10.0.0.9 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
172.16.0.0/30 is subnetted, 4 subnets
R    172.16.7.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    172.16.8.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    172.16.9.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
R    172.16.10.0 [120/6] via 192.168.2.2, 00:00:05, FastEthernet0/0
192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/30 is directly connected, FastEthernet2/1
L    192.168.1.2/32 is directly connected, FastEthernet2/1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/30 is directly connected, FastEthernet0/0
L    192.168.2.1/32 is directly connected, FastEthernet0/0
192.168.3.0/30 is subnetted, 1 subnets
R    192.168.3.0 [120/1] via 192.168.2.2, 00:00:05, FastEthernet0/0
192.168.4.0/30 is subnetted, 1 subnets
R    192.168.4.0 [120/2] via 192.168.2.2, 00:00:05, FastEthernet0/0
192.168.5.0/30 is subnetted, 1 subnets
R    192.168.5.0 [120/2] via 192.168.1.1, 00:00:09, FastEthernet2/1
192.168.6.0/30 is subnetted, 1 subnets
R    192.168.6.0 [120/1] via 192.168.1.1, 00:00:09, FastEthernet2/1
R2#

```

On a pris la table de routage du routeur R2 comme exemple.

❖ APPARITION DES ROUTES

Les routes vers OSPF n'apparaissent plus sur les tables de routage des routeurs RIP car la valeur de métrique par défaut vers les réseaux OSPF est plus élevée que les métriques dans les réseaux RIP or le protocole ne considère pas cette Route comme plus court chemin.

✓ PING de R1 à R8 :

```

R1#
R1#ping 192.168.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 140/178/228 ms
R1#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 208/244/336 ms
R1#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 180/200/232 ms
R1#ping 10.0.0.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 180/216/248 ms
R1#

```

❖ CORRECTION DU PROBLEME

Il faut définir une métrique pour que RIP sache que quelle métrique il doit appliquer parce que c'est la cause du problème (routes ignorées et incomplètes).

Cela permet à RIP d'accepter les routes OSPF et de les propager correctement.

```
R4(config)#router rip
R4(config-router)#version 2
R4(config-router)#no auto-summary
R4(config-router)#network 192.168.3.0
R4(config-router)#network 192.168.4.0
R4(config-router)#redistribute ospf 1 metric 5
R4(config-router)#exit
R4(config)#router ospf 1
R4(config-router)#router-id 4.4.4.4
R4(config-router)#network 172.16.10.0 0.0.0.3 area 0
R4(config-router)#redistribute rip subnets metric 20 metric-type 2
```

• VERIFICATION :

Affichons une des tables de routage des réseaux RIP pour bien vérifier si le problème est résolu :

Test PING :

```
R8#ping 192.168.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 136/152/168 ms
R8#ping 192.168.6.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.6.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 152/167/188 ms
R8#
```

```

R1#ping 10.0.0.8
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 144/152/160 ms
R1#ping 172.16.7.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.7.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 188/222/268 ms
R1#ping 172.16.8.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.16.8.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 184/196/204 ms
R1#

```

Table de routage d'un Routeur :

```

R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override

Gateway of last resort is not set

    10.0.0.0/32 is subnetted, 3 subnets
R       10.0.0.7 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
          [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R       10.0.0.8 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
          [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R       10.0.0.9 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
          [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
    172.16.0.0/30 is subnetted, 4 subnets
R       172.16.7.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
          [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R       172.16.8.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
          [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R       172.16.9.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
          [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
R       172.16.10.0 [120/7] via 192.168.6.1, 00:00:13, FastEthernet2/1
          [120/7] via 192.168.1.2, 00:00:02, FastEthernet0/0
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/30 is directly connected, FastEthernet0/0
L       192.168.1.1/32 is directly connected, FastEthernet0/0
    192.168.2.0/30 is subnetted, 1 subnets
R       192.168.2.0 [120/1] via 192.168.1.2, 00:00:02, FastEthernet0/0
    192.168.3.0/30 is subnetted, 1 subnets
R       192.168.3.0 [120/2] via 192.168.1.2, 00:00:02, FastEthernet0/0
    192.168.4.0/30 is subnetted, 1 subnets
R       192.168.4.0 [120/2] via 192.168.6.1, 00:00:13, FastEthernet2/1
    192.168.5.0/30 is subnetted, 1 subnets
R       192.168.5.0 [120/1] via 192.168.6.1, 00:00:13, FastEthernet2/1
    192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.6.0/30 is directly connected, FastEthernet2/1
L       192.168.6.2/32 is directly connected, FastEthernet2/1
R1#

```

Après la résolution du problème, les communications entre les routeurs doivent être fonctionnelles :

- Toutes les Routes vers les réseaux OSPF apparaissent sur les tables de routage des routeurs RIP.
- Le test de connexion en effectuant des pings est réussi.